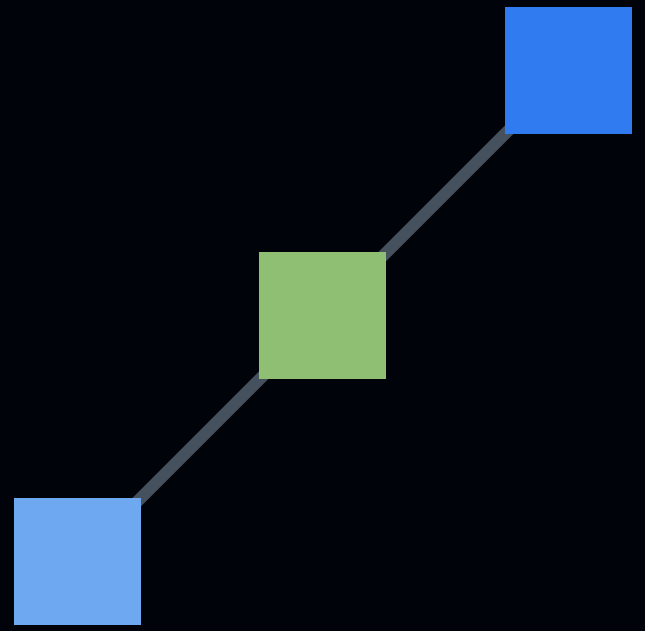
Discrete mathematics · semester project

NextFrame

Graph theory in streaming recommendations

A recommendation engine built as a graph problem over the MovieLens archive: 1,437 titles, 8,498 weighted edges, and every score on screen traceable to a path that produced it.

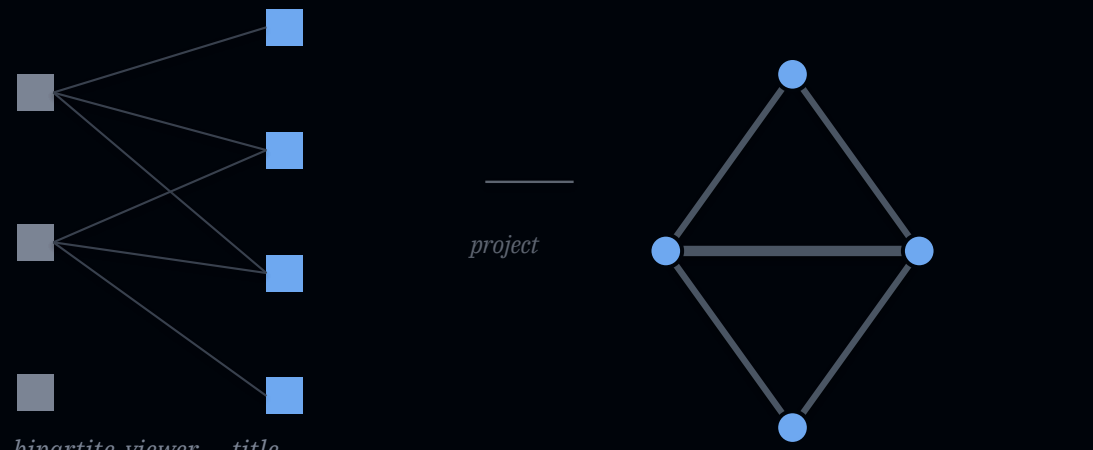


Semester VII · Division B · Batch B2

C039 · C043 · C044 · C046

One graph, built from 100,836 ratings

A rating of 3.5 or more is an edge between a viewer and a title. Two cut-offs keep the graph readable: a title needs ten likes to enter the catalogue, a viewer needs eight catalogue likes to receive a list.



bipartite viewer — title *similarity graph, weighted by Jaccard*

ratings	100,836
likes ≥ 3.5	46,887
titles in catalogue	1,437
viewers	585
weighted edges	8,498
neighbours kept per title	k = 8

The projection is second-order: two titles are adjacent when the same viewers liked both. The weight is the Jaccard index of the two audiences — the share of their combined audience that liked both. The t_2–t_3 edge above is drawn twice as heavy because two viewers support it.

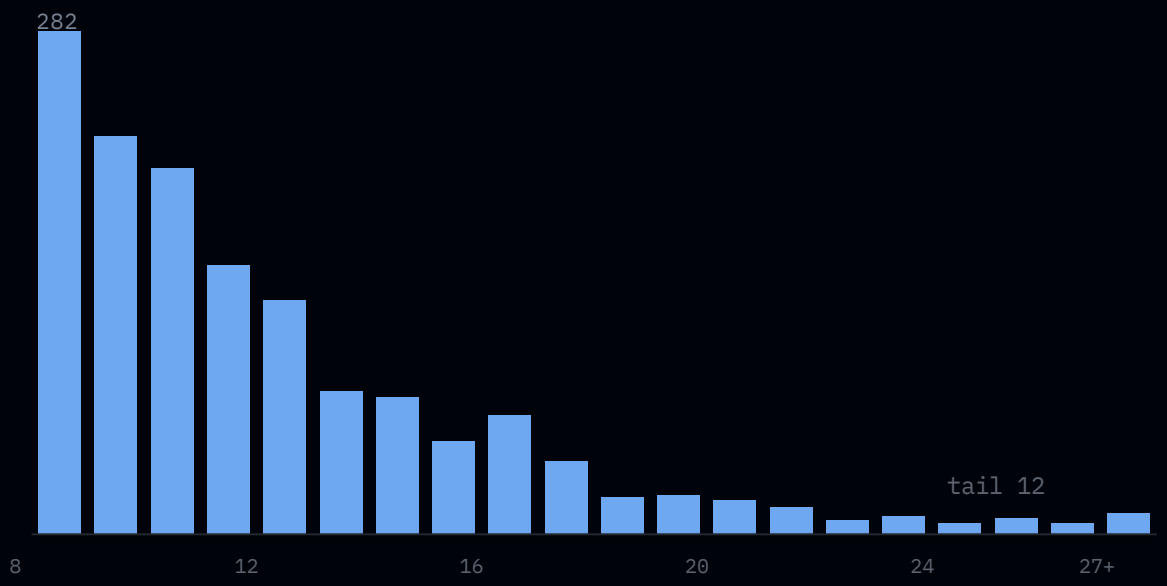
The bundle also carries a recent-releases layer — 68 films from 2025 and 2026 — deliberately outside these counts, because the archive has no ratings for them and inventing an audience size would be a lie.

Two different numbers both get called degree

Audience size d is how many viewers liked a title, and it drives the popularity term in the score. Similarity degree is how many titles share a neighbourhood with it, and it is what the graph topics count. They are different numbers, so the bundle stores them separately.

mean similarity degree	11.83
median	11
maximum	36 · The Sweet Hereafter
sum of degrees	46,887 = 2 × 8,498
edge density	0.0082
mean audience size	32.63

Similarity degree, all 1,437 titles



titles at each degree

The spike at the left is not a property of the data, it is the construction: every title keeps its eight strongest neighbours, so degree 8 is a floor. Above that the tail is genuine — a handful of titles are structurally close to thirty-odd others.

Handshaking lemma: every edge contributes twice, and the ledger balances — 16,996 degree units against 8,498 edges.

One component, no chokepoints, diameter seven

Union–find with path compression and union by rank over all 8,498 edges returns a single component holding every one of the 1,437 titles. The graph has no articulation point and no bridge, so no title is load-bearing and no edge is a chokepoint.

components	1
large component	1,437
articulation points	0
bridges	0
diameter	7
mean path length	3.799
clustering coefficient	0.1676

Viewer 498 — every title by hop distance



own likes	21	1.5%
1 hop	93	6.5%
2 hops	308	21.4%
3 hops	670	46.6%
4+	345	24.0%

The five parts sum to 1,437 — the 21 titles the viewer already liked are at distance zero and are counted before the search begins. Almost half the catalogue is three hops away, and 345 titles sit at four or beyond, outside the three-hop window the recommendation stage searches. That limit is stated in the app rather than hidden.

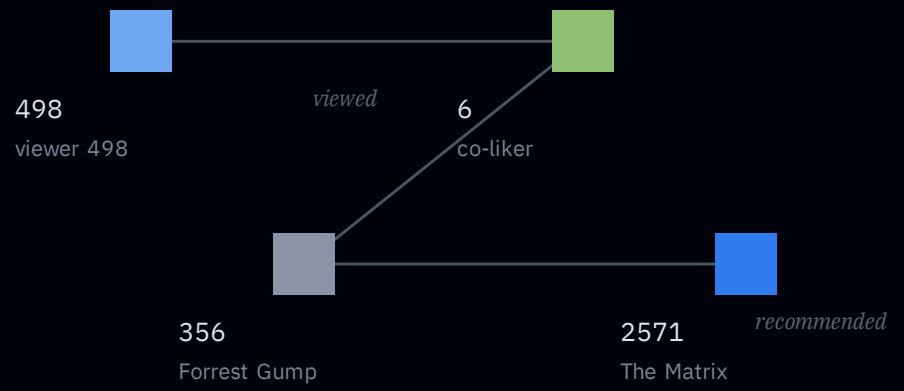
Diameter 7 on 1,437 vertices is the small-world signature: sparse, but the high-audience titles act as hubs.

A recommendation is a path, and the app draws it

A path of length three, alternating between the two sides of the bipartite graph: the viewer, a co-liker, the title the two of them most agree on, and the candidate. The co-liker is the witness, the anchor is the evidence, and the candidate is the answer.

Read the four stops as a walk: viewer, co-liker, anchor, candidate. Close it at the viewer and the same three edges form a 4-cycle in the bipartite graph — exactly what the next slide counts.

one witness path, 498 → The Matrix



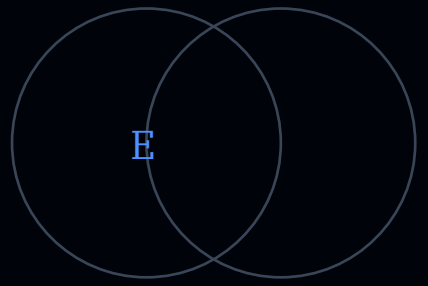
titles 498 and viewer 6 both liked	16
same, for viewers 58 and 45	16 · 14
anchor	Forrest Gump (356)
Jaccard, anchor → candidate	0.409
distinct 4-cycles P	1,124
candidates ranked	1,416

Cycles measure echo, and echo must be divided out

Close the three-hop path at the viewer and it is a 4-cycle. The count of distinct such cycles is the support term P. But a cycle also means the viewer's neighbours already cover the candidate's audience — so P alone rewards echo, and the score has to divide it back out.

triangles	5,487
clustering coefficient	0.1676
girth	3
4-cycles through The Matrix	1,124
novelty 1 / (1 + E)	0.515

echo $E = |A(m) \cap N(u)| \,/\, |A(m)|$



co-likers N(u) audience A(m)

The Matrix 225 / 239	0.941
median viewer	0.055
highest viewer	0.100

Girth 3 says triangles exist, so the projected graph is not bipartite — the projection of a bipartite graph never is. Five and a half thousand triangles at a clustering coefficient of 0.168 is a graph in which two titles that share a neighbour usually share several.

E for The Matrix is 225 / 239: the viewer's immediate neighbourhood already covers 94% of its audience. Its novelty factor is 0.515 — the score halves what the cycle count alone would pay.

The score multiplies three terms, each a graph fact

$$S = (1 + \ln(1 + P)) \times d^{0.35} \times 1 / (1 + E)$$

support
distinct 4-cycles through the candidate, logged because the tail is long

popularity
audience size, damped by α = 0.35 so reach alone does not decide

novelty
the echo share, inverted — agreement already banked pays less

Worked, The Matrix, viewer 498

P cycle count	1,124
1 + ln(1 + P)	8.026
d audience	239
d^0.35	6.799
E echo	0.941
1 / (1 + E)	0.515
S	28.106

1st
rank of 1,416 candidates

3rd
popularity rank, for contrast

8.026 × 6.799 × 0.515 = 28.106. The app prints the three factors under every score, so a viewer can see which one carried it.

Third on raw audience, first on the score: the 0.35 exponent is what stops the most-watched title from winning by default, and the 1,124 cycles are what put it on top.

Euler's theorem says no, and the degrees say why

Two genres are adjacent when a title carries both, so each title with two or more genres contributes one edge — a multigraph of 1,212 edges over 19 vertices. An Eulerian circuit needs every degree even. Ten of the nineteen are odd, so the theorem refuses the walk outright.

genre vertices	19
edges (titles with ≥ 2 genres)	1,212
sum of degrees	2,424 = 2 × 1,212
odd-degree vertices	10
largest degree	Drama, 461

By the handshaking lemma the number of odd vertices must be even, and ten is. That parity is not a coincidence and not a problem — it is the exact quantity the next slide has to spend edges on.

Genre degree · odd vertices in blue

Drama	461	odd
Comedy	358	even
Thriller	54	even
Action	427	odd
Romance	122	even
Adventure	339	odd
Crime	197	odd
Sci-Fi	68	even
Horror	61	odd
Fantasy	42	even
Children	72	even
Animation	99	odd
Mystery	64	even
Documentary	4	even
War	21	odd
Musical	15	odd
Western	9	odd
IMAX	1	odd
Film-Noir	10	even

Euler's condition

connected
yes

all degrees even
no — ten odd

Eulerian circuit
none exists

Ten odd vertices, five pairs, nine duplicated edges

Pair the ten odd vertices by shortest distance in the underlying simple graph, then duplicate every edge along each connecting path. Parity flips at both ends of the path, all degrees become even, and Hierholzer's algorithm returns the circuit.

Odd pairing · distance added

Drama — Action	+1 edge
Adventure — Crime	+1 edge
Horror — Animation	+2 edges
War — Musical	+2 edges
Western — IMAX	+3 edges
total added	+9

edges before	1,212
odd vertices	10 → 5 pairs
edges duplicated	9
edges after	1,221
all degrees even	yes
circuit length	1,221 traversals
output	a playlist that alternates genres

A circuit is a closed walk that uses every edge exactly once — so the playlist never repeats a title, and consecutive titles always share a genre because the walk moves along a genre edge.

the circuit's first eight hops

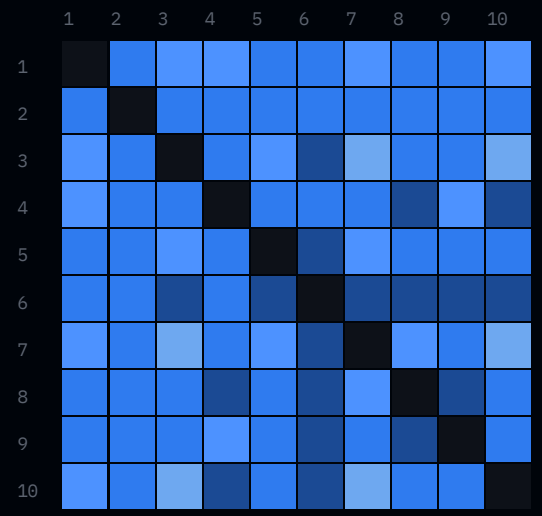
Adventure → Sci-Fi → IMAX → Sci-Fi → Adventure → Western → Adventure → Crime → Adventure

Every hop crosses a genre edge, so the playlist can only move to a neighbouring genre — the alternation is the walk, not a rule added on top. Hierholzer runs in time linear in the edges.

A watch order is a Hamiltonian path over the top ten

Same ten candidates, different question: visit every candidate exactly once, maximising the total similarity along the way. Held–Karp is exact and costs $n^2 \cdot 2^n$ — 10,240 states for n = 10, which is why the list is ten long.

Pairwise Jaccard, the ten candidates



The order

1. Terminator 2: Judgment Day
2. Star Wars: Episode VI - Return of...
3. Star Wars: Episode V - The Empire...
4. Star Wars: Episode IV - A New Hope
5. Raiders of the Lost Ark (Indiana ...
6. Matrix, The
7. Fight Club
8. American Beauty
9. Silence of the Lambs, The
10. Schindler's List

$10 \cdot 2^{10}$

states for n = 10

The DP state is (set visited, last vertex): the best path over that subset ending there. Every state extends by one unvisited vertex, and the full table answers the question exactly rather than approximately.

Ten is a design choice, not a limitation of the idea — the state count doubles with every extra candidate.

optimal total weight	4.212
nearest-neighbour baseline	4.086
the exact method gains	+3.1%
states searched	10,240

The genre graph cannot be drawn without crossings

Drop the multiplicities and keep an edge when two genres co-occur in at least six titles: 19 vertices, 98 edges. Every planar graph with v ≥ 3 obeys e ≤ 3v − 6. The ceiling here is 51, and the graph has 98.

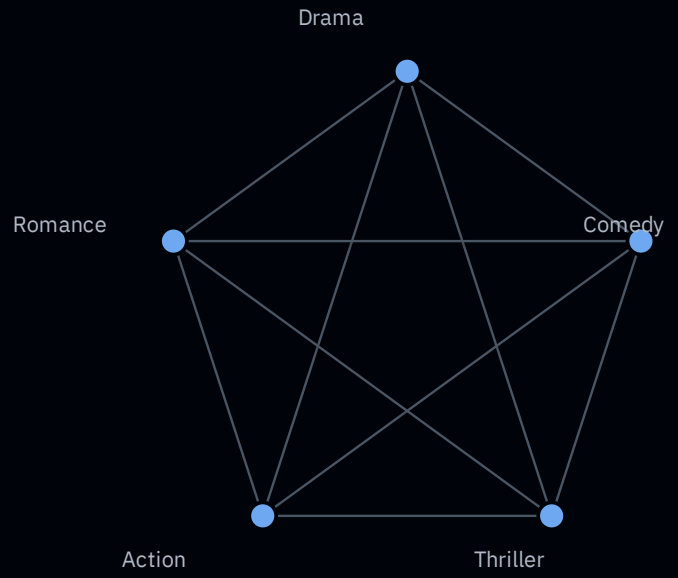
$$e \le 3v - 6$$

$$98 \le 57 - 6 = 51$$

false, so the graph is not planar

vertices / edges	19 / 98
planar ceiling 3v − 6	51
excess	47
Kuratowski	K_5 witness
largest clique	K_9

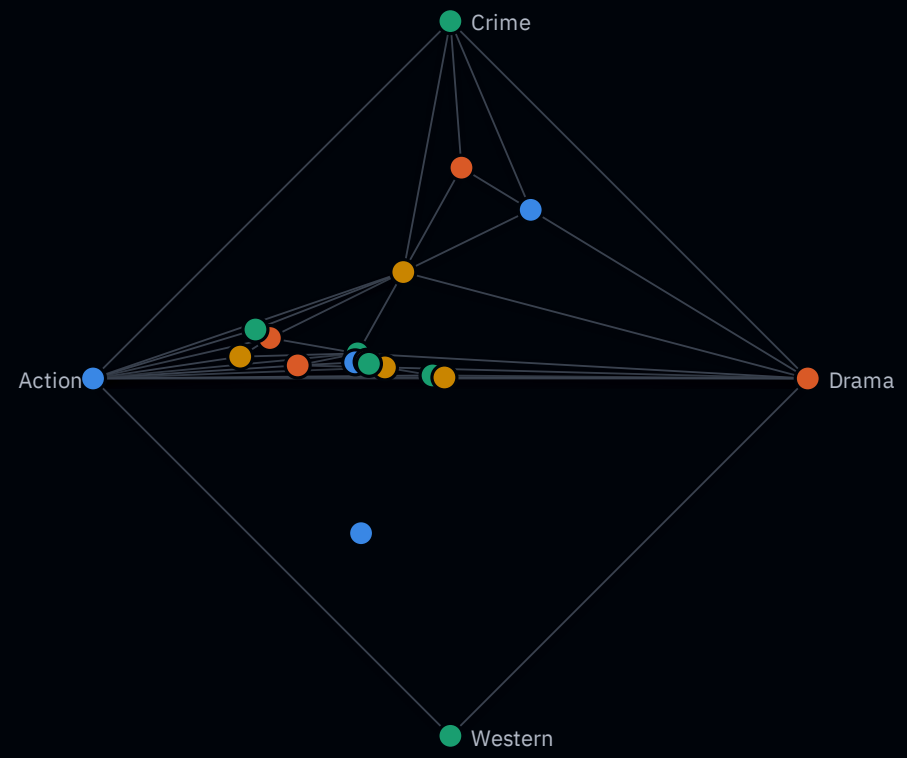
K_5 inside it — the Kuratowski witness



Keep the heavy edges, and four colours suffice

Keep the heaviest edges in descending co-occurrence order and discard any edge that breaks planarity: 47 survive, 51 go. Tutte's barycentric method then places every interior vertex at the average of its neighbours, with the outer face pinned on a circle.

planar subgraph	19 vertices · 47 edges
crossings in the drawing	0
faces F	30
Euler V − E + F	19 − 47 + 30 = 2
colours DSATUR	4 (planar)
largest clique	K_4 → $\chi = 4$ exactly



Tutte drawing · 19 vertices, 47 edges, 0 crossings, 4 colours

The interior crowds because every inner vertex is a barycentre of its neighbours — the drawing is the theorem, not a layout choice.

Colour the list so no two neighbours are adjacent

The conflict graph on one viewer's top ten: two titles are adjacent when they share a genre. A proper colouring is a proof that an ordering exists in which no two similar titles sit next to each other.

vertices	10
adjacency	share a genre
colours DSATUR	7
largest clique	K_7 → $\chi = 7$ exactly
most conflicted	Fight Club, 9
genre graph, planar	4 colours
genre graph, full	9 colours
full-graph clique	K_9

Seven is not just what the heuristic happened to find: the seven titles in the clique all share Action, so any proper colouring needs seven, and DSATUR lands exactly on the bound.

Viewer 498's top ten, in order, by colour

		colour
Matrix, The	Action · Sci-Fi	2
Silence of the Lambs, The	Crime · Horror	3
Star Wars: Episode IV - A New Hope	Action · Adventure	3
Fight Club	Action · Crime	1
Raiders of the Lost Ark (Indiana Jones and the Raiders of the Lost Ark)	Action · Adventure	4
Schindler's List	Drama · War	2
Star Wars: Episode V - The Empire Strikes Back	Action · Adventure	5
Terminator 2: Judgment Day	Action · Sci-Fi	6
American Beauty	Drama · Romance	3
Star Wars: Episode VI - Return of the Jedi	Action · Adventure	7

Seven colours for ten titles: the constraint is tight enough that a greedy pass cannot do better than the clique demands.

Conclusion

Seven topics, one graph, seven answers

Topic	*what it answered*	*the number*
Degree	audience size and similarity degree, kept apart	11.83 mean
Connectivity	one component, no bridge, diameter 7	1,437 reached
Path	three-hop witness paths as support	1,416 candidates
Cycle	4-cycles and echo, divided back out	C 0.168 · 5,487 tri
Eulerian	10 odd vertices, 5 pairs, 9 edges added	1,221 traversals
Hamiltonian	exact max-weight order by Held–Karp	4.212 vs 4.086
Planar + colouring	K_5 blocks it; 4 colours after pruning	$\chi = 4$, tight

Nothing here is illustrated. Every number on these slides is computed from 100,836 ratings and printed in the app, next to the paths that produced it.

Limits

585 viewers is a sample, not a platform.

Recommendations search three hops out; 345 titles sit beyond that.

The archive ends in 2018. Recent releases are carried separately, without audience figures.

C039 degree, connectivity C043 Eulerian, Hamiltonian C044 path, cycle C046 planar, colouring